Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India
(Autonomous College Affiliated to University of Mumbai)

ESE Examination

Max. Marks: 100 March 2024 Duration: 3hr.
Class: T.E. (VI) CSE AIML/DS Course Code: AI306/DS306
Name of the Course: Distributed Computing

Instructions:
(1) All questions are compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Q. No.		Max Marks	CO-BL
Q1 A	Discuss the role of Message Oriented Middleware (MOM) in facilitating communication between distributed applications with neat, labelled diagram. Apply your understanding of MOM limitations to analyse its effectiveness in handling modern communication challenges.	10	1-3
Q1 B	Apply your understanding of communication types by elaborating on each of the following: Persistent asynchronous communication, Persistent synchronous communication, Transient asynchronous communication, Receipt-based transient synchronous communication, Delivery-based transient synchronous communication, and Response-based transient synchronous communication with diagram. Build your response by providing a real-world example for each communication type, detailing how it functions and its significance in the given scenario.	10	1-3
Q2 A	Examine how code migration works in distributed computing systems and how strong mobility helps the gaming platform where a player is in the middle of a game session and decides to switch from their smartphone to their laptop. discuss the pros and cons of strong mobility for the gaming platform, considering things like how easy it is to use, how efficiently it uses resources, and whether it makes the gaming experience better.	10	2-3
Q2 B	Design a load balancing strategy for an e-commerce platform during holiday sales, utilizing round-robin and weighted round-robin algorithms. Explain how this mechanism dynamically allocates resources to manage fluctuating user demand, ensuring scalability and reliability. Discuss challenges and considerations in implementing and managing the chosen load balancing algorithm within the distributed system.	10	3-3

Q3 A	As a system architect, outline how you would leverage replication to enhance the availability and fault tolerance of the e-commerce platform during peak times. Describe the types of replications you would employ and justify your choice based on their advantages and limitations. Additionally, discuss the benefits and challenges associated with replication in ensuring data consistency, fault tolerance and latency in the context of the e-commerce platform.	10	4-3
Q3 B	Discuss Lamport's Logical Clock algorithm. How is the causality capturing limitation fixed by Vector Clocks?	10	3-4
Q4 A	With reference to consistency models, answer the following: 1. Comment on the significance of Data-centric consistency models. 2. Explain distributed data store with a neat diagram. **OR** 1. What is continuous consistency? 2. Discuss Client-centric consistency models.	10	2-2
Q4 B	Describe the following mutual exclusion algorithms in detail with diagrams: 1. Distributed Algorithm 2. Token Ring Algorithm	10	3-2
Q5 A	Discuss Edge-Server Systems as compared with BitTorrent File Sharing Systems. Provide necessary overview diagrams.	10	1-4
Q5 B	Comment on the following providing the necessary concepts and diagrams: 1. The working of GPS 2. Physical clocks on computers	10	3-2